

Underwater Vision-Based Surveillance Using Machine Learning & AI

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REVISED RESEARCH QUESTION

How does underwater image enhancement affect the performance of deep-learning-based object detection across the three supplied underwater datasets?

RESEARCH FOCUS

- Compare original images vs. enhanced images
- Measure enhancement's effect on object-detection performance
- Use a transfer-learning-based YOLO detector as candidate approach
- Evaluate all three supplied datasets — separately, per class structure

Measurable Outputs & Target Definition

OBJECT DETECTION METRICS

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95

IMAGE ENHANCEMENT METRICS (WHERE APPLICABLE)

- PSNR
- SSIM
- UIQM

COMPUTATIONAL PERFORMANCE

- Inference time per image / frame
- Enhancement processing time, if applicable

Main Comparison: *Original* → Detector vs. *Enhanced* → Detector

TARGET DEFINITION

Primary

Detect underwater objects — class, bounding box, confidence score — across all three datasets.

Secondary

Determine whether enhancement improves, has little effect, or degrades detection performance.

DATASET SCOPE — ALL THREE INCLUDED

well.v8i.yolov8	4 classes
DIATAquarium.v4i.yolov8	11 classes
Aquatic Plant.v2i.yolov8	1 class

Evaluated separately — class structures differ across datasets.

Completed Work & Key Findings

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DAY 1 — PROBLEM UNDERSTANDING

- Understood the underwater surveillance problem
- Inspected all three supplied datasets
- Identified classes, splits & YOLO annotations
- Defined pipeline: Input → Enhancement → Detection → Evaluation → Tracking (if applicable)

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DAY 2 — LITERATURE REVIEW

Reviewed 7 closely related papers across enhancement, detection, deep learning, datasets & tracking.

Candidate methods shortlisted:

- White Balance / CLAHE
- Water-Net / FUnIE-GAN
- YOLO-based transfer-learning detector
- IoU / SORT-style tracker (if sequential data)

Problem Understanding

Literature Review

Candidate Methods Identified

Research Question Refined

KEY FINDINGS

- Underwater images suffer color distortion, low contrast, blur & poor visibility
- Project compares original vs. enhanced rather than assuming improvement
- Three datasets have different class structures — evaluated separately
- Literature shows enhancement does not always improve detection accuracy
- YOLO-based detection shortlisted as candidate approach
- Tracking remains conditional on availability of video/sequential data

Blockers, Mentor Confirmation & Next Plan

POINTS REQUIRING MENTOR CONFIRMATION

Dataset Configuration

Confirm the validation-path issue in Well & DIAT Aquarium data.yaml files (pointing to test/images instead of valid/images)

A) All 3 datasets will be used separately

Evaluation Metrics

Confirm which detection and enhancement metrics are mandatory

A) Yes, this is the main comparison. Since literature shows enhancement doesn't always improve detection, the project tests both versions on the same detector instead of assuming enhancement helps.

Enhancement Comparison

Confirm whether the main experiment compares Original → Detection vs. Enhanced → Detection

A) Precision, Recall, mAP@0.5, and mAP@0.5:0.95 are mandatory for detection. For enhancement, PSNR/SSIM need reference (clean) images which the datasets may not have — so UIQM (no-reference) is proposed as the fallback if reference images aren't available.

Tracking

Confirm whether video/sequential data is available and whether tracking is required

A) NO Tracking

NEXT PLAN

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Day 4 — Dataset Readiness

- Define schema, labels, targets & metadata
- Identify missing-data, imbalance, leakage risks
- Define correct train/valid/test split
- Prepare dataset-risk checklist

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Day 5 — Pipeline & Experiment Planning

- Design reproducible workflow
- Define baseline, advanced & project-specific experiment
- Finalize tentative evaluation criteria
- Create experiment log & repo structure

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Day 6 — Methodology & Mentor Review 2

- Complete literature, methodology & data-readiness package
- Present preprocessing, modelling & evaluation plan
- Confirm split strategy, metrics & constraints
- Prepare Methodology V1 & updated plan